

WEEK 1 PROJECT REPORT

Superstore Sales Data Import, Power Query & Data Cleaning

1. Introduction

This project focuses on importing and preparing the Superstore Sales dataset using Microsoft Power BI. The main purpose of Week 1 is to understand how raw sales data can be imported, inspected, cleaned, and transformed before analysis.

Power Query Editor was used to inspect the dataset and perform the required data-cleaning and transformation activities. The Order Date column was used to create additional date information such as Day and Year.

The Applied Steps section was reviewed to understand the sequence of transformations performed. Finally, the completed changes were applied using Close & Apply, making the prepared dataset ready for further analysis and visualization.

2. Objectives

- To import the Superstore Sales dataset into Power BI.
- To inspect the imported data.
- To clean and standardize the dataset.
- To check the Order Date data type.
- To create a Day column from Order Date.
- To create a Year column from Order Date.
- To review the Applied Steps.
- To apply the final changes using Close & Apply.

3. Tools and Technologies Used

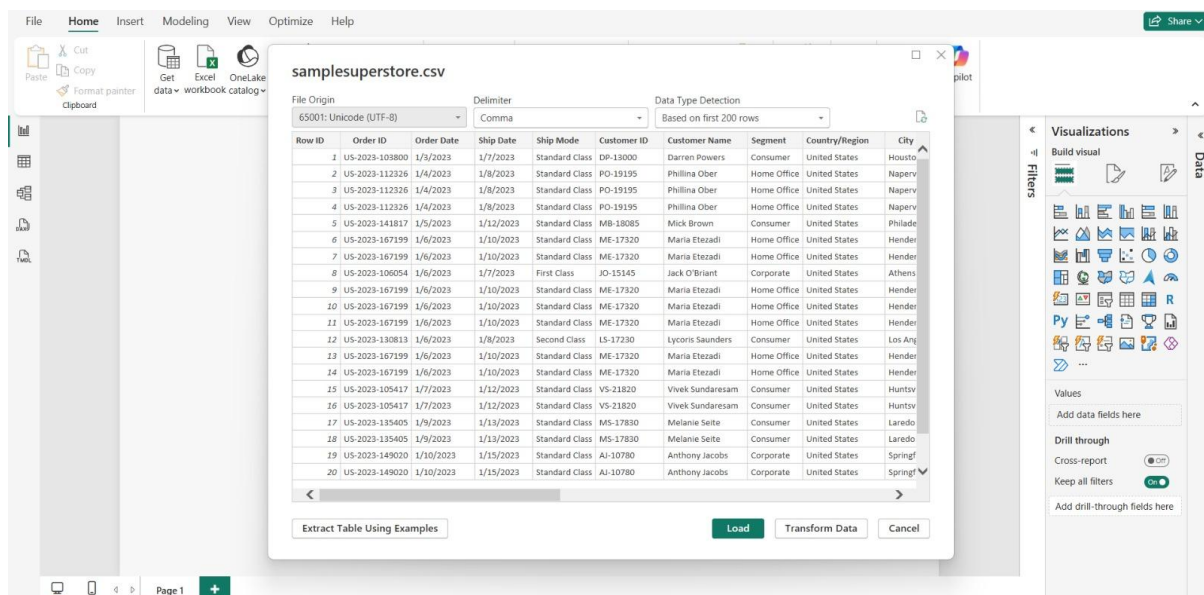
Tool	Purpose
Microsoft Power BI Desktop	Data import and analysis
Power Query Editor	Data cleaning and transformation
Power Query M	Creating custom columns
Superstore Sales CSV	Source dataset

Tool

Purpose

4.Step 1 – Importing the Superstore Sales Dataset

The Superstore Sales dataset was imported into Power BI using the Text/CSV option. The CSV file was previewed to verify that the required columns and records were available before performing data transformations.



Step 2 – Initial Data Inspection

After importing the dataset, the data was opened in Power Query Editor for initial inspection. The columns, records, and values were reviewed to understand the structure and contents of the dataset before further transformation.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City	State/Province
12	US-2023-130813	06 January 2023	08 January 2023	SECOND CLASS	LS-17230	LYCORIS SAUNDERS	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA
57	US-2023-148614	20 January 2023	25 January 2023	STANDARD CLASS	MV-17485	MARK VAN HUFF	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA
122	US-2023-164903	20 February 2023	24 February 2023	STANDARD CLASS	SR-20740	STEVEN ROELLE	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA
331	US-2023-102652	06 April 2023	12 April 2023	STANDARD CLASS	AY-10555	ANDY YOTOV	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
332	US-2023-102652	06 April 2023	12 April 2023	STANDARD CLASS	AY-10555	ANDY YOTOV	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
370	US-2023-120740	15 April 2023	15 April 2023	SAME DAY	PS-18970	PAUL STEVENSON	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA
373	US-2023-110849	18 April 2023	23 April 2023	STANDARD CLASS	JL-15835	JOHN LEE	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA
375	US-2023-110849	18 April 2023	23 April 2023	STANDARD CLASS	JL-15835	JOHN LEE	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA
377	US-2023-110849	18 April 2023	23 April 2023	STANDARD CLASS	JL-15835	JOHN LEE	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA
723	US-2023-114125	09 July 2023	13 July 2023	STANDARD CLASS	GH-14410	GARY HANSEN	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA
727	US-2023-114125	09 July 2023	13 July 2023	STANDARD CLASS	GH-14410	GARY HANSEN	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA
1409	US-2023-101931	28 October 2023	31 October 2023	FIRST CLASS	TS-21370	TODD SUMRALL	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
1410	US-2023-101931	28 October 2023	31 October 2023	FIRST CLASS	TS-21370	TODD SUMRALL	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
1411	US-2023-101931	28 October 2023	31 October 2023	FIRST CLASS	TS-21370	TODD SUMRALL	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
1422	US-2023-106439	31 October 2023	04 November 2023	STANDARD CLASS	GG-14650	GREG GUTHRIE	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
1427	US-2023-106439	31 October 2023	04 November 2023	STANDARD CLASS	GG-14650	GREG GUTHRIE	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
1428	US-2023-106439	31 October 2023	04 November 2023	STANDARD CLASS	GG-14650	GREG GUTHRIE	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
1430	US-2023-106439	31 October 2023	04 November 2023	STANDARD CLASS	GG-14650	GREG GUTHRIE	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
1431	US-2023-106439	31 October 2023	04 November 2023	STANDARD CLASS	GG-14650	GREG GUTHRIE	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
1446	US-2023-168305	01 November 2023	08 November 2023	STANDARD CLASS	PL-18925	PAUL LUCAS	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA
1449	US-2023-168305	01 November 2023	08 November 2023	STANDARD CLASS	PL-18925	PAUL LUCAS	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA
1468	US-2023-153619	03 November 2023	03 November 2023	SAME DAY	TC-21535	TRACY COLLINS	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA
1816	US-2023-138737	07 December 2023	10 December 2023	FIRST CLASS	FP-14320	FRANK PREIS	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA
1859	US-2023-113579	13 December 2023	15 December 2023	SECOND CLASS	KD-16345	KATHERINE DUCICH	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA
2288	US-2024-138219	23 March 2024	30 March 2024	STANDARD CLASS	BP-11095	BART PISTOLE	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA
3056	US-2024-121188	28 August 2024	04 September 2024	STANDARD CLASS	CB-12025	CASSANDRA BRANDOW	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA

Step 3 – Checking the Order Date Data Type

The Order Date column was checked in Power Query to ensure that it was recognized as a Date data type. Correctly identifying the date type is important for performing date-based transformations such as extracting the Day and Year.

The screenshot shows the Power Query Editor interface. The main area displays a table with columns: Row ID, Order ID, Order, Ship Mode, Customer ID, and Customer Name. The 'Order' column is selected, and a context menu is open with the 'Change Type' option chosen. The 'Change Type' submenu is visible, showing various data types. 'Date' is highlighted, indicating the selection of the Date data type for the 'Order' column. The right sidebar shows the 'PROPERTIES' and 'APPLIED STEPS' panes. The 'APPLIED STEPS' pane shows a list of steps, including 'Inserted Day', which is the current step being applied.

Step 4 – Creating the Day Column

The Order Date column was used to extract the Day value. Power Query's Transform options were used to create a separate Day column from the Order Date field.

The screenshot shows the Power Query Editor interface. The 'Transform' tab is active, and the 'Day' option under 'Text Transforms' is highlighted in the context menu. The background table shows columns: Row ID, Order ID, Order Date, Ship Mode, Customer ID, and Customer Name. The 'Order Date' column is being transformed to extract the day of the month.

Row ID	Order ID	Order Date	Ship Mode	Customer ID	Customer Name
1	US-2023-105800	11-01-2023	Standard Class	DP-13000	Darren Powers
2	US-2023-112326	14-01-2023	Standard Class	PO-19195	Phillina Ober
3	US-2023-112326	18-01-2023	Standard Class	PO-19195	Phillina Ober
4	US-2023-112326	16-01-2023	Standard Class	PO-19195	Phillina Ober
5	US-2023-141817	15-01-2023	Standard Class	MB-18085	Mick Brown
6	US-2023-167199	15-01-2023	Standard Class	ME-17320	Maria Etezad
7	US-2023-167199	14-01-2023	Standard Class	ME-17320	Maria Etezad
8	US-2023-106054	15-01-2023	Standard Class	E-17320	Jack O'Brian
9	US-2023-167199	15-01-2023	Standard Class	E-17320	Maria Etezad
10	US-2023-167199	15-01-2023	Standard Class	E-17320	Maria Etezad
11	US-2023-130813	15-01-2023	Standard Class	E-17320	Maria Etezad
12	US-2023-167199	15-01-2023	Standard Class	E-17320	Maria Etezad
13	US-2023-167199	15-01-2023	Standard Class	E-17320	Maria Etezad
14	US-2023-167199	15-01-2023	Standard Class	E-17320	Maria Etezad
15	US-2023-105417	15-01-2023	Standard Class	ME-17320	Maria Etezad
16	US-2023-105417	15-01-2023	Standard Class	V5-21820	Vivek Sundaresam
17	US-2023-135405	15-01-2023	Standard Class	V5-21820	Vivek Sundaresam
18	US-2023-135405	15-01-2023	Standard Class	MS-17830	Melanie Seite
19	US-2023-149020	15-01-2023	Standard Class	MS-17830	Melanie Seite
20	US-2023-149020	15-01-2023	Standard Class	AJ-10780	Anthony Jacobs
21	US-2023-130092	11-01-2023	First Class	SV-20365	Seth Vernon
22	US-2023-157147	13-01-2023	Standard Class	BD-11605	Brian Dahlen
23	US-2023-109232	13-01-2023	Second Class	ND-18370	Natalie DeCherney
24	US-2023-157147	13-01-2023	Standard Class	BD-11605	Brian Dahlen
25	US-2023-118192	13-01-2023	Standard Class	MM-17920	Michael Moore
26	US-2023-162775	13-01-2023	Standard Class	CS-12250	Chris Selesnick

Step 6 – Creating the Year Column

A Custom Column was created to extract the year from the Order Date column using Power Query M.

The screenshot shows the 'Custom Column' dialog box. The 'New column name' is 'Year'. The 'Custom column formula' is: `=Table.AddColumn(#"Changed Type1", "Year", each Date.Year([Order Date]), Int64.Type)`. The 'Available columns' list includes Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, and Customer Name. The 'OK' button is highlighted.

Row ID	Order ID	Order Date	Ship Mode	Customer ID	Customer Name
1	US-2023-105800	11-01-2023	Standard Class	DP-13000	Darren Powers
2	US-2023-112326	14-01-2023	Standard Class	PO-19195	Phillina Ober
3	US-2023-112326	18-01-2023	Standard Class	PO-19195	Phillina Ober
4	US-2023-112326	16-01-2023	Standard Class	PO-19195	Phillina Ober
5	US-2023-141817	15-01-2023	Standard Class	MB-18085	Mick Brown
6	US-2023-167199	15-01-2023	Standard Class	ME-17320	Maria Etezad
7	US-2023-167199	14-01-2023	Standard Class	ME-17320	Maria Etezad
8	US-2023-106054	15-01-2023	Standard Class	E-17320	Jack O'Brian
9	US-2023-167199	15-01-2023	Standard Class	E-17320	Maria Etezad
10	US-2023-167199	15-01-2023	Standard Class	E-17320	Maria Etezad
11	US-2023-130813	15-01-2023	Standard Class	E-17320	Maria Etezad
12	US-2023-167199	15-01-2023	Standard Class	E-17320	Maria Etezad
13	US-2023-167199	15-01-2023	Standard Class	E-17320	Maria Etezad
14	US-2023-167199	15-01-2023	Standard Class	E-17320	Maria Etezad
15	US-2023-105417	15-01-2023	Standard Class	ME-17320	Maria Etezad
16	US-2023-105417	15-01-2023	Standard Class	V5-21820	Vivek Sundaresam
17	US-2023-135405	15-01-2023	Standard Class	V5-21820	Vivek Sundaresam
18	US-2023-135405	15-01-2023	Standard Class	MS-17830	Melanie Seite
19	US-2023-149020	15-01-2023	Standard Class	MS-17830	Melanie Seite
20	US-2023-149020	15-01-2023	Standard Class	AJ-10780	Anthony Jacobs
21	US-2023-130092	11-01-2023	First Class	SV-20365	Seth Vernon
22	US-2023-157147	13-01-2023	Standard Class	BD-11605	Brian Dahlen
23	US-2023-109232	13-01-2023	Second Class	ND-18370	Natalie DeCherney

Step 7 – Final Data Preparation and Close & Apply

After completing the required transformations, the dataset was reviewed and the changes were applied using Close & Apply. The prepared data was then loaded into Power BI for further analysis.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name
1	US-2023-103800	03-01-2023	07-01-2023	Standard Class	DP-13000	Darren Powers
2	US-2023-112326	04-01-2023	08-01-2023	Standard Class	PO-19195	Phillina Ober
3	US-2023-112326	04-01-2023	08-01-2023	Standard Class	PO-19195	Phillina Ober
4	US-2023-112326	04-01-2023	08-01-2023	Standard Class	PO-19195	Phillina Ober
5	US-2023-141817	05-01-2023	12-01-2023	Standard Class	MB-18085	Mick Brown
6	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etessadi
7	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etessadi
8	US-2023-190654	06-01-2023	07-01-2023	First Class	JO-15145	Jack O'Brian
9	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etessadi
10	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etessadi
11	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etessadi
12	US-2023-130813	06-01-2023	08-01-2023	Second Class	LS-17230	Lycoris Saunders
13	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etessadi
14	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etessadi
15	US-2023-105417	07-01-2023	12-01-2023	Standard Class	VS-21820	Vivek Sundaresam
16	US-2023-105417	07-01-2023	12-01-2023	Standard Class	VS-21820	Vivek Sundaresam
17	US-2023-135405	09-01-2023	13-01-2023	Standard Class	MS-17830	Melanie Sette
18	US-2023-135405	09-01-2023	13-01-2023	Standard Class	MS-17830	Melanie Sette
19	US-2023-149020	10-01-2023	15-01-2023	Standard Class	AJ-10780	Anthony Jacobs
20	US-2023-149020	10-01-2023	15-01-2023	Standard Class	AJ-10780	Anthony Jacobs
21	US-2023-130992	11-01-2023	14-01-2023	First Class	SV-20365	Seth Vernon
22	US-2023-157147	13-01-2023	18-01-2023	Standard Class	BD-11605	Brian Dahlien
23	US-2023-109232	13-01-2023	16-01-2023	Second Class	ND-18170	Natalie DeCherney
24	US-2023-157147	13-01-2023	18-01-2023	Standard Class	BD-11605	Brian Dahlien
25	US-2023-118192	13-01-2023	18-01-2023	Standard Class	MM-17920	Michael Moore
26	US-2023-162775	13-01-2023	15-01-2023	Second Class	CS-12250	Chris Selesnick
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5. Project Workflow

Superstore Sales CSV Dataset



Import Data into Power BI



Open Power Query Editor



Inspect the Dataset



Check Order Date Data Type



Create Day Column



Review Applied Steps



Create Year Column

↓

Review Final Data

↓

Close & Apply

6.Deliverables

- Superstore Sales dataset imported into Power BI.
- Dataset inspected using Power Query.
- Order Date data type checked.
- Day column created from Order Date.
- Year column created from Order Date.
- Applied Steps reviewed.
- Final changes applied using Close & Apply.

6. Learning Outcomes

- Import CSV data into Power BI.
- Use Power Query Editor.
- Inspect and clean data.
- Check and manage data types.
- Perform date transformations.
- Create Day and Year columns.
- Use Power Query M.
- Understand Applied Steps.
- Apply transformations using Close & Apply.

7.Conclusion

The Week 1 project provided practical experience in importing, inspecting, cleaning, and transforming the Superstore Sales dataset using Power BI and Power Query.

The Order Date column was prepared and used to create Day and Year columns. The Applied Steps were reviewed to understand the transformation process, and the final changes were applied using Close & Apply.

